

Jemin Lho

📍 Queens, NY ✉ jeminlho.engineering@gmail.com ☎ 347-684-3828 🔗 [linkedin.com/in/jeminlho/](https://www.linkedin.com/in/jeminlho/)

EDUCATION

Stony Brook University | *Bachelor of Engineering in Mechanical Engineering*

August 2021 – May 2025

- **Awards:** Cum Laude, Dean's List **GPA: 3.67/4.0**
- **Relevant Coursework:** Mechatronics, Applied Stress Analysis, Thermodynamics II, Fluid Mechanics, Thermal System Design, Heat Transfer, Material Science, Machine Element Design

SKILLS & CERTIFICATIONS

Certifications: FE Mechanical, E.I.T. (No. 097811), CompTIA A+, SolidWorks CSWP (In Progress)

Software: Fusion 360, SolidWorks, Abaqus CAE, AutoCAD, ANSYS

Programming: MATLAB, Python, Arduino, LabVIEW, LaTeX

Manufacturing: Milling, Lathing, 3D Printing, Laser Cutting, MIG Welding, Root Cause Analysis

EXPERIENCE

Assistant Technician | *New York Network Solution Inc.*

October 2025 – Present

- Diagnosed and repaired computer hardware through systematic troubleshooting and root-cause analysis, resolving system reliability issues to implement cost-effective solutions.
- Installed private security camera systems under supervision of a Senior Network Engineer involving physical mounting, alignment, and wiring, reinforcing principles of system integration and field installation.
- Expanded manageable parts inventory and **tripled** component sales while tracking stock levels and coordinating replacements to support timely repairs and installations.

Undergraduate Research Assistant | *The Soft Flyers Group*

September 2023 – December 2024

- Led an independent research project to gather performance data on bistable wing systems for aerial robots with a primary focus on validating MATLAB simulations and improving experimental consistency.
- Designed and fabricated 3D-printed support joints and brackets for experimental setups, achieving an estimated fatigue life of 10,000 cycles, leveraging FEA in Fusion 360 to streamlined wing structural testing and construction.
- Developed a Python OpenCV-based program for real-time positional data collection on multiple points of interest at a rate of 30 frames per second, utilizing Euclidean distance calculation and camera lens-distortion correction algorithms.
- Documented results and processes in a detailed 50-page technical deliverable, coinciding with weekly deadlines, while collaborating with aspiring PhD students and assistant professors to prepare for future publication.

Powertrain Driveshaft and Axles Systems | *Stony Brook Motorsports Team*

August 2023 – May 2024

- Collaborated with a multidisciplinary team to assemble and iterate upon a safe, high-performance off-road vehicle meeting specified constraints through optimal axle placements and safety features.
- Manufactured driveshaft hoops and powertrain guards in compliance with Baja SAE competition technical and safety standards.
- Updated SolidWorks CAD models for vehicle chassis and created engineering drawings using appropriate GD&T processes. Designs were imported into ANSYS FEA simulations to analyze failure scenarios during possible vehicle collisions and cornering cases.
- Celebrated the team's top 10 finish in the 2024 Baja SAE Williamsport endurance race with a top 25 overall design evaluated vehicle.

PROJECTS

Industrial Plastic Shredder | *University Senior Design Project*

[[Video](#) & [Project Files](#)]

- Collaborated with university faculty and three student peers to iteratively design, develop, and manufacture a machine to crush 3D-printed materials for an economical recycling process, increasing filament yield by 20% per roll with multiple possible recycling cycles.
- Prepared a 250-page technical documentation including mechanical drawings, performance calculations, product specifications, and FEA simulations to innovate against market-available designs with budget constraints of 10% that of market rates.
- Experimentally tested and documented machine performance through audio-vibrational analysis, temperature analysis, and failure tests.
- Successfully shredded 4 kilograms of unused ABS 3D-prints at 70% stress limits that reprocessed into 22 kilograms of new filament.

Japanese Arcade Capsule Machine | *Graduate Mechatronics Academic Project*

[[Video](#) & [Project Files](#)]

- Designed and prototyped a capsule dispensing arcade machine with three graduate and undergraduate students with features including a SolidWorks-modeled chassis and closed-loop logic, reinforcing mechatronic principles.

Budget Quadcopter | *Stony Brook American Institute of Aeronautics & Astronautics (AIAA)*

- Designed a bluetooth-controlled quadcopter with Proportional-Integral-Derivative control and a lightweight 3D-printed chassis whose fully assembled weight was 34% of the maximum propeller thrust force.
- Optimized PID controller parameters in flight control software, reducing rise time to <60 milliseconds and overshoot to 7% for enhanced quadcopter vibrational stability and responsiveness.